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AI School Iteration 2.2

Linear Regression

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01. What is Regression?

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01. What Is Regression?

Regression = Predicting a continuous number as output

Examples:

- Predict house prices
- Predict tomorrow's temperature
- Predict a student's exam score
- Predict monthly sales

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02. Simple Linear Regression

02. Simple Linear Regression

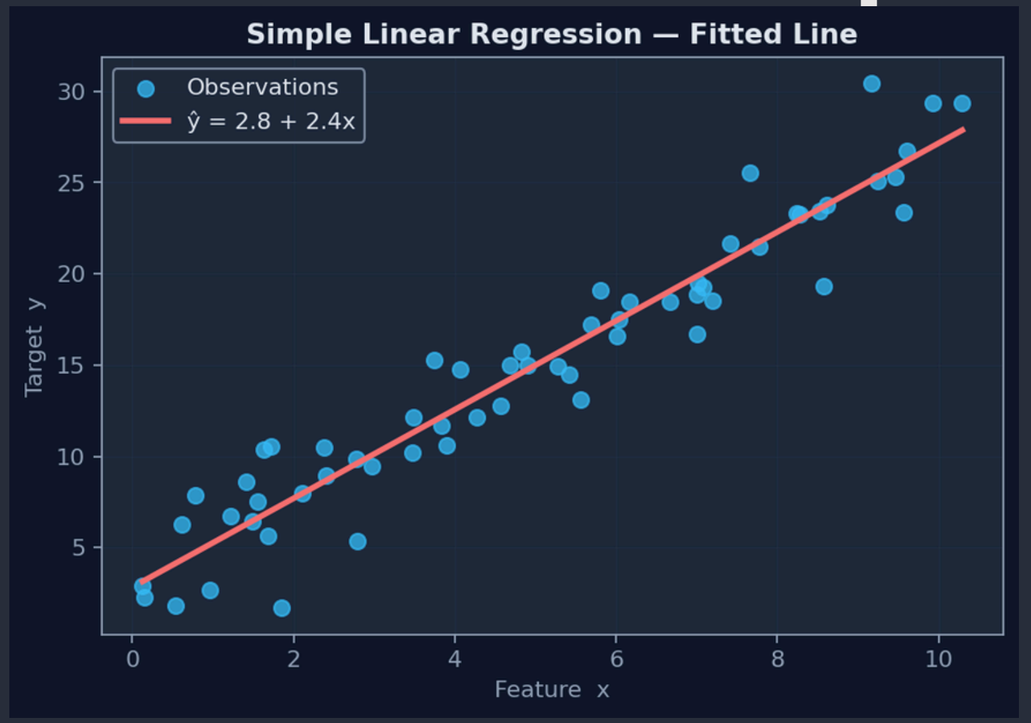
Key idea: Find **a straight line** that best describes the relationship between one input and one output.

$$\hat{y} = \beta_0 + \beta_1 x$$

| $\hat{y}$: Predicted value (dependent variable)
| x: Input feature (independent variable)
| β_0: Intercept – where the line crosses the y-axis
| β_1: Slope – how much $\hat{y}$ changes per unit of x

Why simple? Only **one** independent variable
Why linear? The equation is a **linear** function of x

02. Simple Linear Regression



There are infinite possible lines. We need a way to pick the *best* one.

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03. Parameter Optimization

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03. Residuals



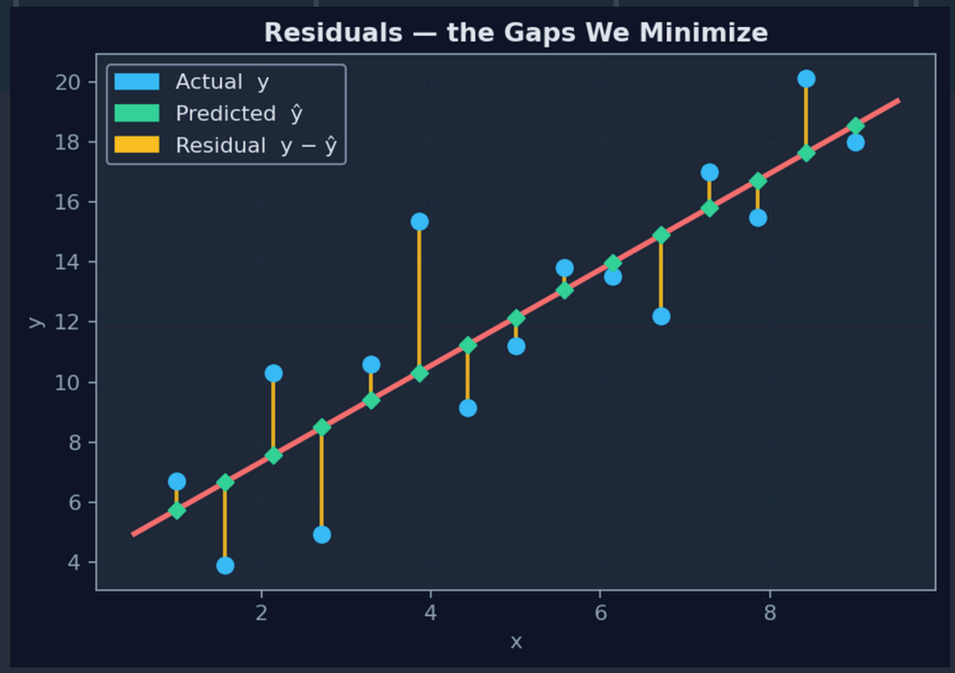
A residual is the gap between the actual value and what the model predicted.

$$residual = y - \hat{y}$$

Rule of thumb

In supervised learning, the goal is always to **minimize the gap** between predicted and actual values.

03. Loss function (MSE)



$$\text{Mean Squared Error (MSE)} = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$$

- We square residuals so negatives don't cancel out positives
- We want this number **as small as possible**

=> Residual Sum of Squares minimization → leads to the best fit line

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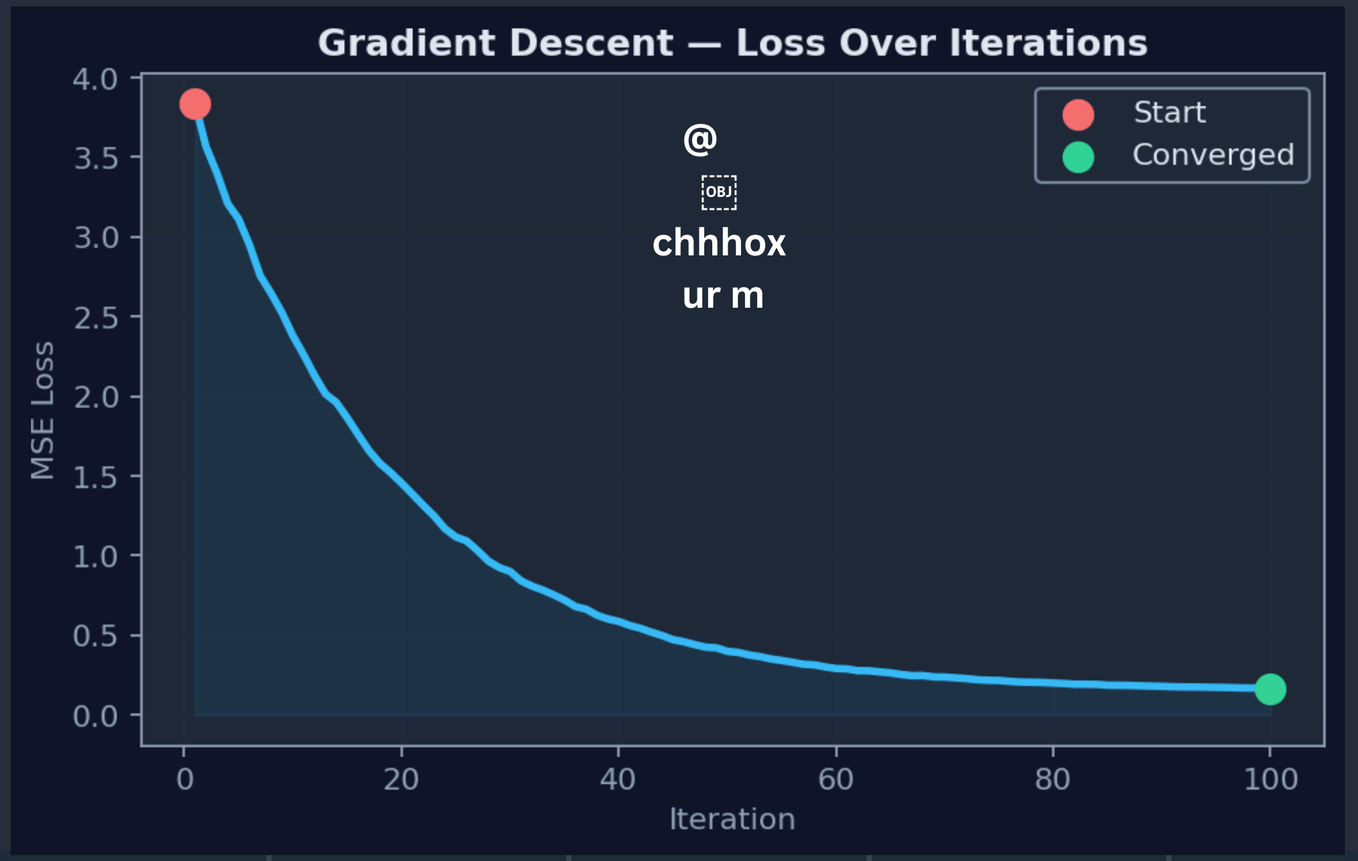
03. How Do We Minimize the Loss?

Ordinary Least Squares (OLS)	Gradient Descent
- A mathematical formula that solves for β directly - Closed-form, exact, fast for small datasets $\beta = (X^T X)^{-1} X^T y$ No iterations needed Expensive when data is very large	- An iterative algorithm — takes small steps downhill on the loss curve - **Used in most ML frameworks under the hood** Scales to large datasets Requires choosing a **learning rate**

Both methods find the same answer. OLS is exact, gradient descent is approximate but scalable.

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03. How Do We Minimize the Loss?



Learning Curve

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03. Gradient Descent

Step 1 — Forward Pass: compute the prediction

Given the current weights, run the model forward and compute ŷ:

$$\hat{y} = W_0 + W_1X_1 + W_2X_2 + \ldots + W_nX_n$$

Then compute the loss:

$$J(w) = MSE = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$$

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03. Gradient Descent

Step 2 — Measure each parameter's contribution to the loss

For every weight w_i we need to know two things:

- Direction: does this value of w_i drive the loss up or down?
- Magnitude: by how much?

This is exactly what the gradient tells us:

$$\nabla w_i = \partial J / \partial w_i$$

> The gradient is the same concept as a derivative, the slope of the loss with respect to w_i at the current point.
> A positive gradient means w_i is pushing the loss up.
> A negative gradient means w_i is pushing the loss down.

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03. Gradient Descent

Step 3 — Backward Pass: update every parameter

We want to reduce the loss, so we move each weight in the opposite direction of its gradient:

$$w_i(t+1) = w_i(t) - \eta \cdot \nabla w_i$$

Gradient sign	Meaning	What we do
$\nabla w_i > 0$	w_i drives loss up	Subtract → reduce w_i
$\nabla w_i < 0$	w_i drives loss down	Subtracting a negative → increase w_i

The learning rate η controls the step size, how aggressively we update.
Too large → overshoot the minimum. Too small → painfully slow convergence.

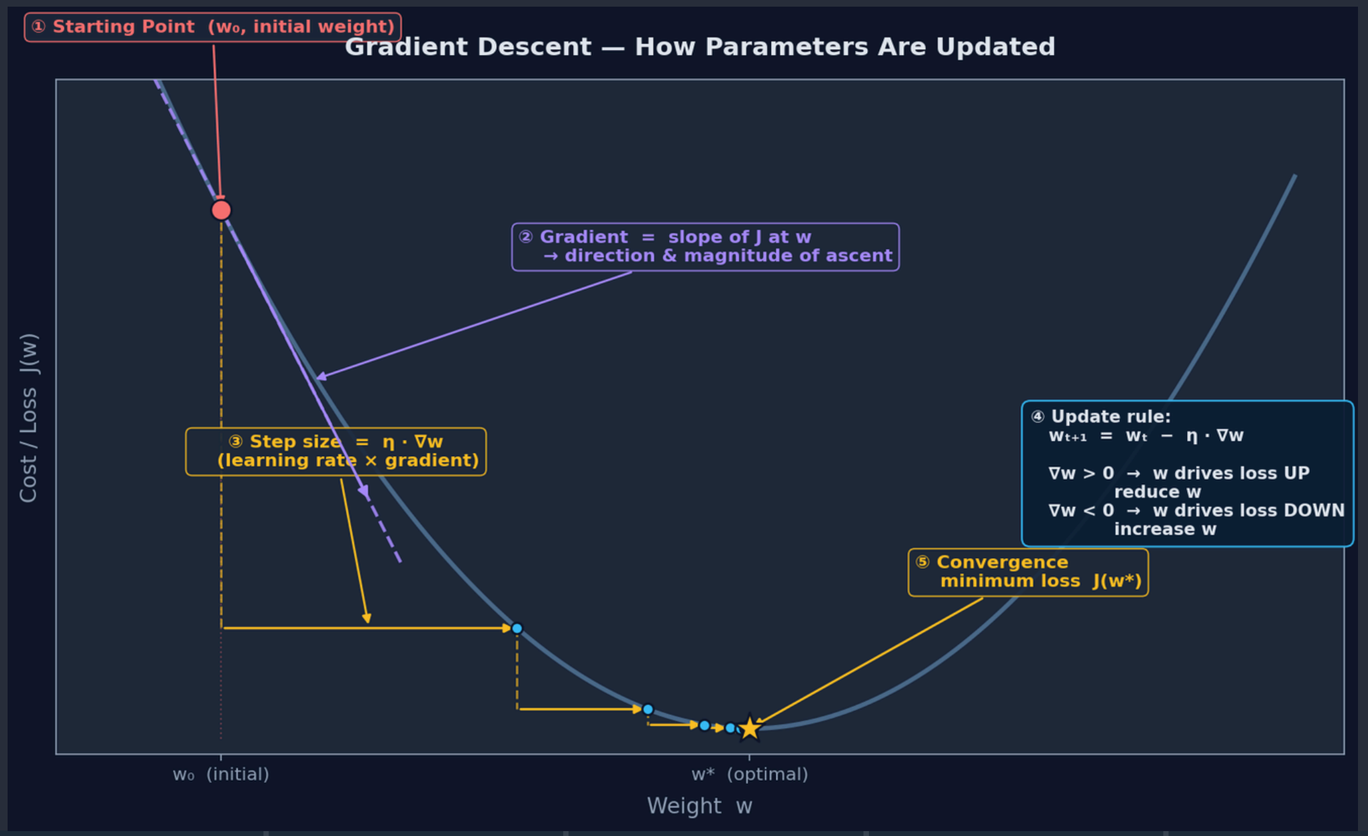
03. Gradient Descent

Step 4 — Repeat until convergence

We repeat steps 1–3 for many iterations. Each pass:
- Loss decreases (ideally)
- Parameters get closer to their optimal values
- Steps get smaller as we approach the minimum (gradient shrinks near flat regions)

We stop when the loss stops improving significantly → **convergence**.

03. Gradient Descent



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04. Multiple Linear Regression

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04. Multiple Linear Regression

Key idea: Same concept, more input features.

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

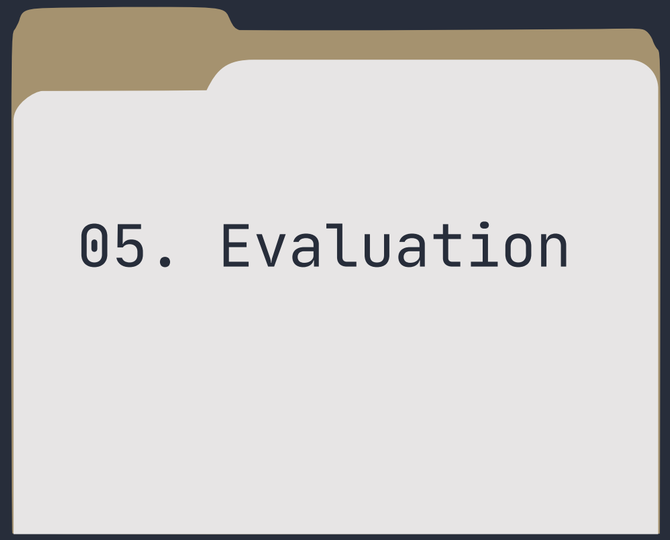
Example:
Predicting house price using:

- x_1 = size (m^2)
- x_2 = number of rooms
- x_3 = distance from city center

Interpreting coefficients:
$\beta_1 = 200$ means: for every 1 m^2 increase in size, the predicted price increases by 200 (holding all other features constant).

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05. Evaluation: RMSE

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

- Same unit as the target variable (interpretable!)
- Penalizes large errors more heavily (due to squaring)
- "On average, my predictions are off by X units"

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05. Evaluation: MAE

Mean Absolute Error (MAE)

$$MAE = (1/n) \Sigma |y_i - \hat{y}_i|$$

- Average absolute error: simple and robust
- Less sensitive to outliers than RMSE
- "On average, I'm off by X, treating all errors equally"

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05. Evaluation: R^2

R-squared

$$R^2 = 1 - (SS_{res} / SS_{tot})$$

Residual Sum of Squares: $SS_{res} = \Sigma (y_i - \hat{y}_i)^2$
Total Sum of Squares: $SS_{tot} = \Sigma (y_i - \bar{y})^2$

- Measures how well the model explains the variance in the data
- Ranges from **0 to 1 (higher is better)**, can go negative if the model is worse than a flat line
- "My model explains X% of the variation in the outcome"

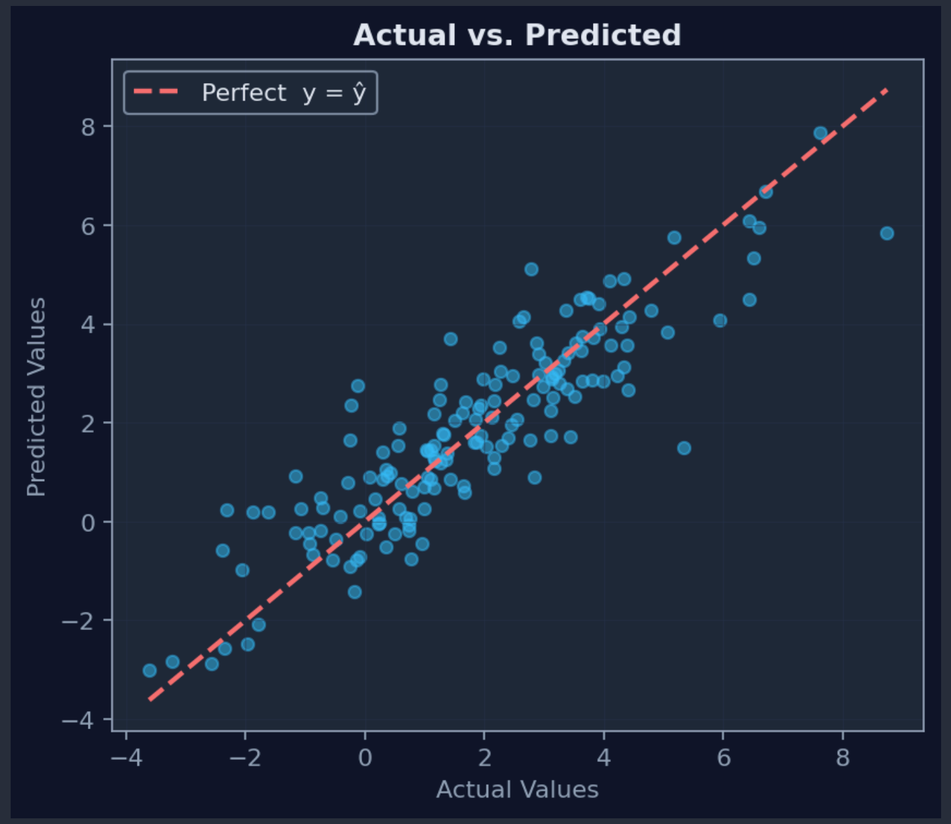
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06. Visual Evaluation

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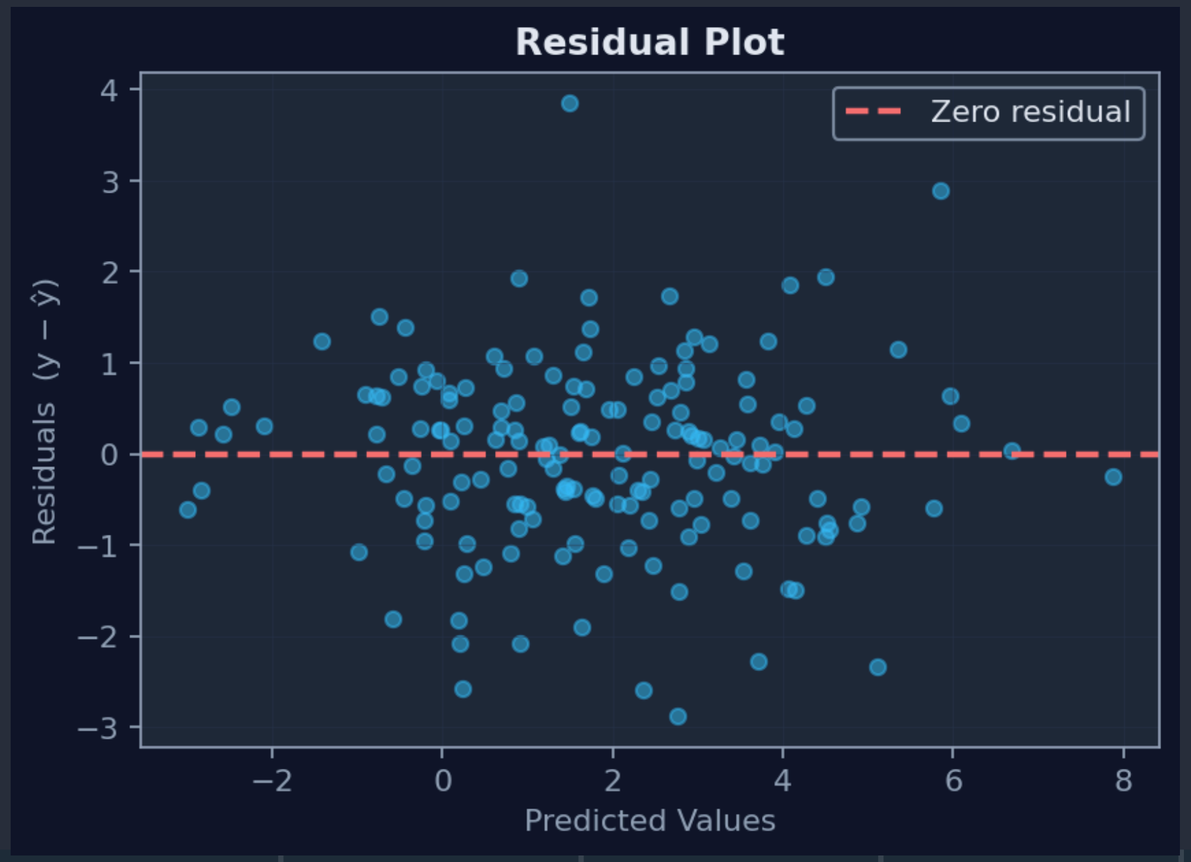
06. Actual vs. Predicted Plot



- x-axis: actual values, y-axis: predicted values
- **A perfect model → all points on the diagonal line y = x**
- Clusters away from the line reveal systematic errors

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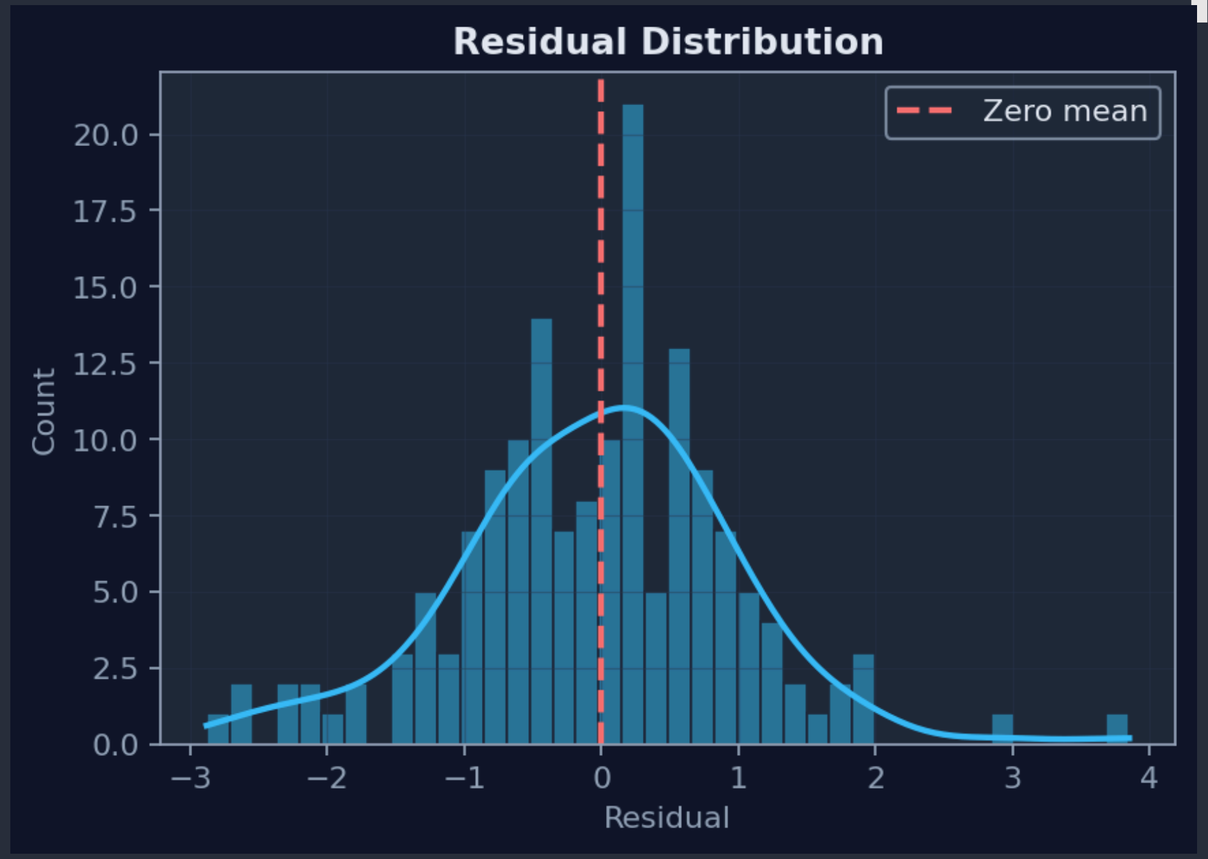
06. Residual Plot



- x-axis: predicted values, y-axis: residuals
- Good model → residuals scattered randomly around 0
- Patterns in this plot = your model is missing something

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06. Residual Distribution Plot

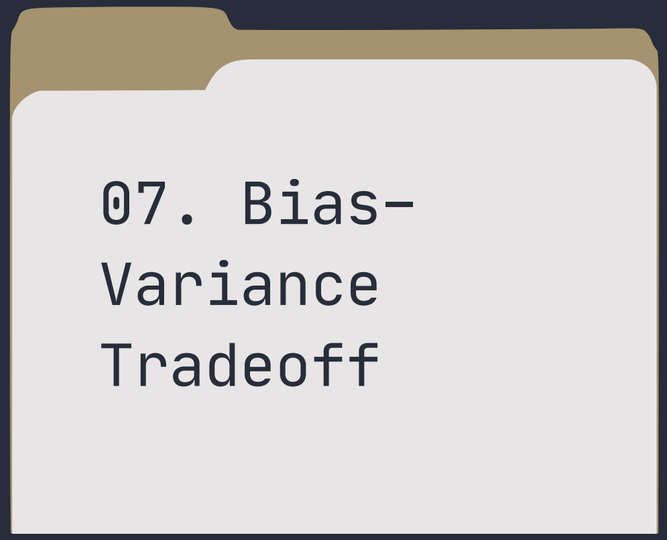


- Histogram or KDE of residuals
- Should look roughly normal, centered at 0
- Skewed distribution → model has a bias

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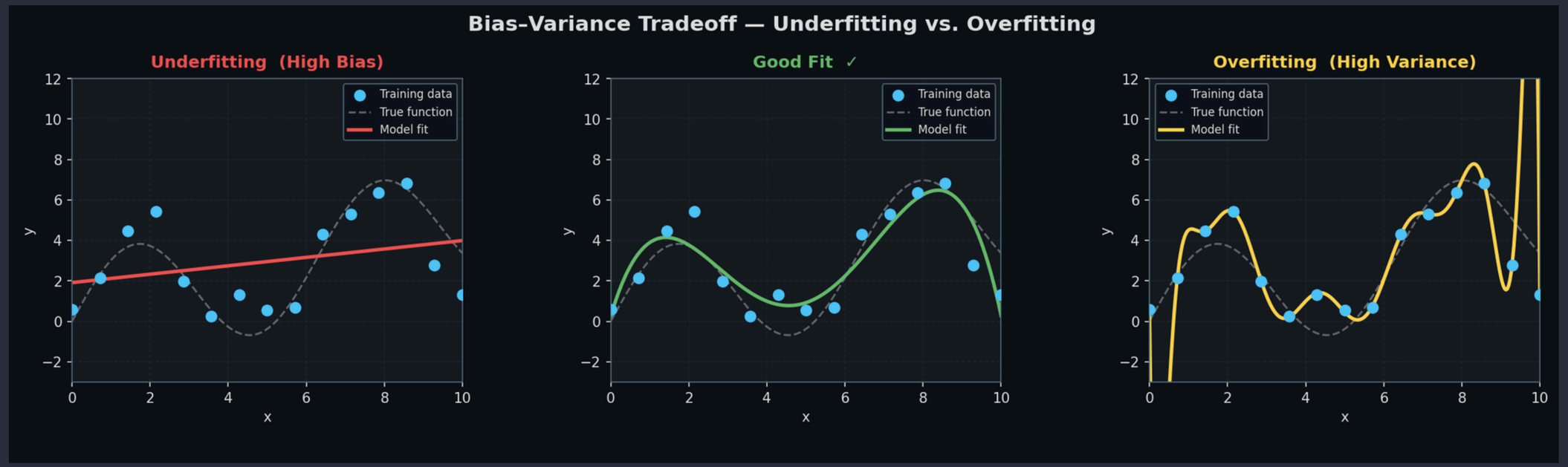
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Numbers lie. Plots don't. Always visualize before concluding.



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07. Bias-Variance Trade-off



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07. Bias-Variance Trade-off

Bias => Underfitting	Variance => Overfitting
• The model is **too simple** • **H**igh bias → **underfitting:** misses the real pattern even on training data	• The model is **too complex** • High variance → **overfitting**: memorizes noise, fails on new data
Bad Performance on Both Train & Test (or Val) Sets Training error ≈ Test error (both high) → Underfitting → **add complexity**	**Good Performance on Train X Bad Performance on Val/Test** Training error low, Test error high → Overfitting → **simplify or regularize / add data**

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08. Regularization: Ridge & Lasso

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08. Regularization

Problem: When a linear regression model overfits, its coefficients grow very large.

Solution: Add a penalty term to the loss function that discourages large coefficients.

08. Ridge (L2)

$$\text{Loss} = \text{MSE} + \lambda \cdot \Sigma \beta_i^2$$

- Penalizes the square of each coefficient
- Shrinks all coefficients toward zero, but never exactly to zero
- Great when many features contribute a little
- Controlled by λ (lambda): larger λ → stronger shrinkage

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08. Lasso Regression (L1)

$$\text{Loss} = \text{MSE} + \lambda \cdot \Sigma \, |\beta_i|$$

- Penalizes the absolute value of each coefficient
- Can shrink coefficients all the way to zero → built-in feature selection
- Great when only a few features actually matter

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Any Questions ?

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Thank you for your attention

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